

Acute Kidney Injury — Pre / Intra / Post Framework

Abrupt ↑Cr · eGFR is invalid during AKI · history + UA + FeNa + renal US sort the three buckets · rule out obstruction early (reversible)

Define it (KDIGO 2012)

Any one	Threshold
↑ Cr (48 h)	≥ 0.3 mg/dL
↑ Cr (7 d)	≥ 1.5 × baseline
Urine output	< 0.5 mL/kg/h

eGFR is meaningless in AKI — the equations assume a steady-state GFR. Characterize as non-oliguric / oliguric / anuric.

3 buckets & how to tell them apart

Bucket	Clinches it
PRE-renal ↓ EABV	History + bland UA + no hydro + FeNa <1% (if oliguric)
INTRA-renal	"Active" sediment; FeNa >1–2% while oliguric; normal US
POST-renal	Hydronephrosis on renal US

On diuretics, FeNa is unreliable → use **FeUrea** <35% for pre-renal.

Pre-renal causes

- ↓ EABV: hypovolemia, sepsis, heart failure (cardiorenal), cirrhosis (hepatorenal)
- Vasogenic: afferent constriction (**NSAIDs, CNIs**) / efferent dilation (**ACEi, ARB**)
- Altered vasculature: renal vein thrombosis, renal venous HTN (CHF), intra-abdominal hypertension

Intra-renal — by compartment

Site	Causes / clues
Glomerular	GN (immune-complex, pauci-immune, anti-GBM; crescentic=RPGN) → dysmorphic RBCs, RBC casts, proteinuria
Tubular	ATN (ischemic or toxic incl. contrast) → muddy-brown casts; tubular obstruction (myoglobin, light chains; crystals: acyclovir, ethylene glycol, MTX, uric acid)
Interstitial	AIN — drugs (beta-lactams, NSAIDs, PPIs, immunotherapy), infection, Sjögren's/sarcoid → pyuria, WBC casts
Vascular	MAHA (TTP/HUS, scleroderma renal crisis, malignant HTN, HELLP); cholesterol emboli

Post-renal — obstruction

- **Urethra**: stricture, BPH · **Bladder**: clot, malignancy, neurogenic · **Ureter**: calculi, int/ext malignancy or LAN, retroperitoneal fibrosis
- **Unilateral obstruction ≠ AKI** unless it is the single functioning kidney — needs **bilateral** obstruction
- Diagnose with renal US (hydronephrosis); rule out **early** — easily reversible

Manage by bucket / Do / AVOID

- **Pre-renal** → restore perfusion (volume, hemodynamics)
- **Intra-renal** → treat cause; **remove nephrotoxins** (NSAIDs, contrast, ACEi/ARB); supportive
- **Post-renal** → relieve obstruction (catheter, stent/nephrostomy)
- **AVOID** trusting eGFR or a single FeNa on diuretics
- **AVOID** overhydrating the anuric ATN patient — maintain euolemia
- **AVOID** skipping the renal US — missed obstruction is missed cure